GGG-W-340a
March 6, 1963

SUPERSEDING
Int. Fed. Spec. GGG-W-00340(Navy-Ships)
February 16, 1961 and
Fed. Spec. GGG-P-171c (In Part) (see 6.3)
January 8, 1960

FEDERAL SPECIFICATION

WIRE TWISTER PLIER (WITH SIDE CUTTER)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers wire twister pliers which are used primarily for twisting safety wires on critical equipment on aircraft, automotive equipment, and other similar application.

1.1.1 *Federal specification coverage.* Federal specifications do not include all types and classes of the commodity indicated by the titles of the specifications, but cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Type and sizes.* The pliers shall be of one type only and of the following sizes as specified (see 6.1):

9-inch nominal length.
12-inch nominal length.

2. APPLICABLE SPECIFICATIONS, AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools, Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5120

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 **Construction.** Wire twister pliers shall be of lap joint, plier-type construction and shall incorporate a device which, when actuated, will cause the tool to revolve about its longitudinal axis. Each half of the basic plier shall include a jaw, handle and joint section which shall be forged integral. Handle extensions, when used, shall be forged and shall be permanently affixed to the handle of the basic plier by welding. The halves of the plier shall be permanently joined together with a through rivet which shall act as the pivot or fulcrum point of the tool. A locking device shall be included which, when engaged, shall hold the handles in a closed position in such a manner as to cause the jaws to firmly and securely grip and hold strands of lockwire.

3.2 **Material.** Materials used in the construction of wire twister pliers shall be as hereinafter specified and shall be of uniform quality, free of injurious defects. Strength and other physical properties of all material shall be adequate for the specific purpose and shall meet or exceed the applicable requirements, including test requirements, listed herein.

3.3 **Surface finish.** The surface of the wire twister plier shall be free from pits, nodules, burrs, excess oxide scale, and other surface defects. Surfaces shall be finished natural, machined, ground and polished, or a combination of these finishes.

3.4 **Coatings.** Unless otherwise specified (see 6.1), the surfaces of wire twisters, except ground and polished surfaces, shall have one or a combination of the following coatings:

(*a*) Oil treated.
(*b*) Chemically produced oxide coating, oil treated.
(*c*) Chemically produced phosphate coating, oil treated.
(*d*) Electro-deposited metallic coating.

3.5 **Identification marking.** Each wire twister plier shall be stamped or otherwise permanently marked in a plain and legible manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.6 **Handles.** For purposes of this specification, handles shall be considered to include the joint section and the handle extension (if used). Handles shall be forged from steel of such quality that when properly heat treated shall have a hardness of not less than 35 nor more than 50 on the Rockwell "C" scale when tested in accordance with 4.3.1 and will comply with the requirements hereinafter specified. The handles shall be shaped so as to provide a safe, comfortable handgrip.

3.7 **Jaws.** For purposes of this specification, the jaw shall be considered to include the entire jaw section of the plier extending from the joint section to the tip end. The jaws shall be provided with an integral wire cutter adjacent to the joint section. Jaws shall be forged from steel of such quality that when properly heat treated shall have a hardness of not less than 45 nor more than 60 on the Rockwell "C" scale when tested in accordance with 4.3.1 except that area which is immediately adjacent to the cutting edge of the wire cutter which shall possess a hardness of not less than 55 nor more than 60 on the Rockwell "C" scale when similarly tested. The gripping portion of the jaws shall be scored or serrated in such a manner that such scoring or serrations will actually improve the grip of the jaws. The scores or serrations shall not be so sharp or so deep that damage to lockwire would result. A wire clearance opening shall be provided between the two jaws and shall be located between the gripping area and the cutter areas of the jaws. This opening shall be of a size and design which will permit the user to swiftly and easily deflect doubled strands of wire out of the side of the jaw and away from the

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:39Z
Check the source to verify that this is the currentversion before use.

cutting edges when performing continuous (multiple) lock wiring operations. Unless otherwise specified (see 6.1) jaw tips shall be the manufacturer's standard commercial design (rounded, slightly tapered, or diagonally pointed to one side opposite the cutting edges) but shall not include the designs known commercially as "duck billed" or "needle nosed". The integral wire cutter shall meet the requirements and tests specified in 4.3.3.

3.8 Joint rivet. The ends of the rivet shall be finished flush with the adjacent sides of the tool. The rivet shall be of proper size and shall be so fitted in the pliers that there will be no excessive movement or other indications of looseness between the tool halves and the rivet when in any operating position. The rivet shall possess a hardness of not less than 20 nor more than 50 on the Rockwell "C" scale when tested in accordance with 4.3.1.

3.9 *Spiral mechanism.* The spiral mechanism shall be capable of producing a close, uniform twist in safety wire without binding of or damage to the mechanism. This mechanism shall consist essentially of a spiral rod and housing, a spiral rod guide, and a pull knob. This mechanism shall be mounted between the handles of the tool and shall be permanently affixed to one handle by either riveting or welding in such a position that the longitudinal axis of the spiral rod shall correspond to the longitudinal axis of the plier. When rivets are used, they shall be of solid type. The design of this mechanism shall be such that when the spiral rod is withdrawn from the housing, the entire tool will be caused to revolve about its longitudinal axis. The total travel of the spiral rod during each withdrawal shall cause the tool to make the following minimum number of complete revolutions:

(*a*) 9-inch size—3 revolutions.

(*b*) 12-inch size—4 revolutions.

3.9.1 *Spiral rod and housing.* The spiral rod employed shall be made of steel and shall have multiple right-hand flutes or grooves. A steel housing shall be provided which shall completely encase the spiral rod when in the retracted position in order to protect the rod from dirt or damage. A suitable stop shall be provided to limit the travel of the spiral rod and prevent accidental separation of the rod from the housing. Suitable means shall be provided to retain the spiral rod in the retracted position within the housing when the tool is held in a vertical position with the handle ends pointed down.

3.9.2 *Pull knob.* The spiral rod shall be equipped with a suitable metal pull knob for actuating the spiral mechanism. That portion of the knob designed for gripping with the thumb and forefinger shall be not less than 11/16 inch in diameter and if cylindrical shall be knurled to prevent slipping from the fingers in operation. In the event the knob is flared to prevent slippage of the fingers, knurling will not be required. The design of the knob shall permit a comfortable, adequate grip by the operator.

3.9.3 *Spiral rod guide.* A device shall be provided which will act as a guide for the spiral rod. The design shall be such that it will cause the wire twister to revolve when the spiral rod is withdrawn. If a sleeve type guide is used, it shall be of bronze or other suitable bearing material. The bearing surface of the sleeve shall have a surface finish no rougher than 63 r.m.s. and shall have not less than 1-inch long bearing surface. If a caged ball bearing and pawl type guide is used, the device shall employ at least 2 hardened and ground steel balls. The balls shall be at least 1/8-inch in diameter. In either case, the design of the guide and the material used shall be such as to provide long life and ease of operation to the spiral mechanism.

3.9.4 *Locking device.* The wire twister shall be equipped with a locking device sufficiently durable to withstand maximum usage. The device shall be so designed that when the plier jaws are closed, the handles

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

must be further compressed in order to engage the lock. The resultant grip between the plier jaws in this locked position shall be such as to firmly and securely grip any size lockwire from 0.010 to 0.060 inch. Release of the locking device shall be accomplished automatically by a spring loaded device when the handles are compressed with one hand while holding the pliers in the same hand. The locking device shall not interfere with the user's hands when performing lock wiring operations.

3.10 Wire twister pliers shall conform to table I for the size specified (see 6.1) and shall be similar to figure 1.

3.11 **Workmanship.** Workmanship shall be first class in every respect. All surfaces shall be free of burrs, fins, nodules or projections of any kind. Forged parts shall be free of laps, cracks and pits. Welded areas shall be neatly and smoothly blended with the adjoining members.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

TABLE I.—*Dimensions of pliers, wire twister, with side cutter*

Nominal size	A ±1/2	B ±1/8	C +3/32 −1/16	D ±1/16	E ±1/4	F	Weight maximum
Inches	*Inches*	*Inches*	*Inch*	*Inch*	*Inches*	*Inches*	*Ounces*
9	8-7/8	1-1/2	7/16	13/16	1-1/2	10-1/4±1/4	13
12	10-3/16	1-1/2	7/16	13/16	1-3/4	11-3/8±1/2	15

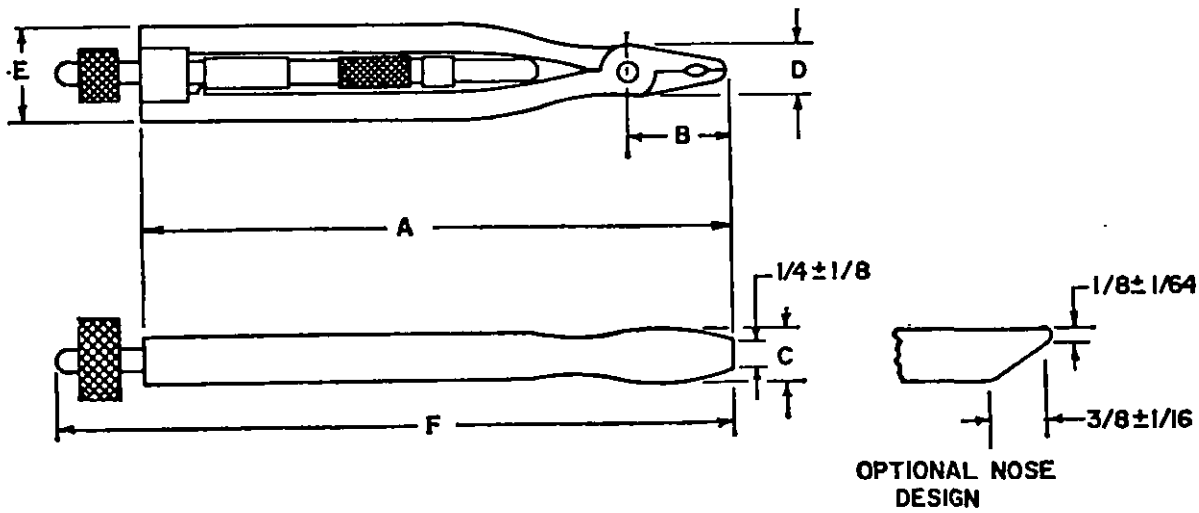


FIGURE I.—Pliers, wire twister, with side cutter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

4.1.1 *Materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and examined in accordance with referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Inspection.

4.2.1 *Lot.* All pliers of the same size shall be considered a lot for purposes of sampling and inspection.

4.2.2 *Examination.* A random sample of pliers shall be selected from each lot for examination in accordance with MIL-STD-105 at inspection level III for lots of 40 and under, inspection level II for lots of 41 to 300, and inspection level I for lots of 301 and over. The acceptable quality level (AQL) shall be 1.5 percent defective. Examination shall be conducted as specified in table II. Any plier in the sample containing one or more defects shall be cause for rejection of the defective plier, and if the number of defective pliers in any sample exceeds the acceptance number for the sample, this shall cause rejection of the lot represented by the sample.

TABLE II.—*Classification of defects in accordance with MIL-STD-105*

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Size not as specified.
102	Material not forged steel.
103	Excessive side play of the jaws or joints.
104	Jaws cannot be opened by the action of one hand.
105	Gripping surfaces defective or not properly scored.
Major (cont'd)	
106	Cutting edges defective, evidence of nicks, burrs, or "wire edges", not aligned, or not provided with adequate clearance or recess, or fail to make contact throughout the entire length when in the closed position.
107	Handles not smooth; evidence of fins, burrs, rough edges, or sharp corners.
108	Coating (if required) missing, incomplete, defective, or not as specified.
109	Weight of the wire twister pliers exceeds the maximum allowable.
Minor:	
201	Jaw dimensions not within specified tolerances.
202	Overall length not within specified tolerances.
203	Distance between handles when in the closed position not within specified tolerance.
204	Jaw thickness and width not uniformly tapered from the joint.
205	Marking, manufacturer's name or trademark missing, illegible, incorrect, or not permanently marked.

4.2.3 *Tests.* A random sample of pliers shall be selected from each lot in accordance with MIL-STD-105 at inspection level L-4. The acceptable quality level shall be 1.5 percent defective. Samples shall be tested as specified in 4.4 for the applicable size.

4.3 Test procedures.

4.3.1 *Hardness.* The jaw hardness of the wire twister pliers shall be of the mean value of three or more determinations taken in an area adjacent to the gripping surfaces and cutting edges. Samples ordinarily shall not be destroyed by fracturing in order to obtain suitable jaw surfaces for test. If reliable readings cannot be obtained on the finished wire twister plier because of difficulty in supporting the test sample and

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:39Z
Check the source to verify that this is the current version before use.

maintaining the test surface level at the point of indentation, a suitable fixture shall be employed for clamping or properly supporting the sample under test. The jaws of wire twister pliers furnished with a coated finish shall be prepared for the hardness test by grinding only sufficiently to insure accurate readings. Hardness determinations for handles shall be taken approximately midway between the rivet and the end of the handles. For hardness determination, a lot may consist of handles, jaws, or rivets, as applicable, heat treated as a batch or under the same process and temperature control. Hardness determination may be performed at any appropriate intermediate station prior to finishing. The sample unit for hardness determinations may be one handle, jaw, rivet, or finished plier, as applicable. All hardness determinations shall be made on a Rockwell tester, using the Rockwell "C" scale or a suitable equivalent scale on the Rockwell testing machines.

4.3.2 *Wire twisting and gripping tests.* Two types of safety wire shall be used for the wire twisting tests: 0.020 inch diameter single strand annealed copper wire and 0.051 inch diameter annealed nickel-chromium-iron alloy (Inconel) wire. Sample wires shall be cut to 18 inch lengths, plus or minus 1/4 inch. The spiral mechanism of the pliers shall be lubricated with a light machine oil for purposes of the test. Examination of pliers shall show no visible signs of cracks, undue wear, faulty operation, deformation, looseness, misalignment of plier parts, or any other harmful defect, after conducting the tests specified in 4.3.2.2 and 4.3.2.3.

4.3.2.1 *Test equipment.* The equipment required for performing the wire twisting tests shall consist of two weights. Each weight shall have an eyebolt, "U" bolt, "J" bolt, or other suitable coupler securely attached to the weight so that a test wire may be looped through the coupler to lift the weight in an approximately balanced position. The eyebolt or other coupler shall have a 1/2 inch to 5/8 inch nominal cross section diameter at the point contacted by the test wire when it is used to lift the weight. The weight may be designed so that it can be prevented from rotating on its vertical axis during the wire twisting test, provided it is not restrained from being lifted. The size of the weight for performing the test specified in 4.3.2.2 shall be 35 pounds, plus or minus 1/2 pound. For the test specified in 4.3.2.3, the weight shall be seven pounds, plus or minus 1/2 pound.

4.3.2.2 *Large wire twisting test.* The 0.051 inch diameter Inconel wire shall be attached to the 35 pound weight by passing one end through the eyebolt or other coupler being used, and then gripping both free ends of the wire with the jaws of the pliers. The ends of the wire shall be inserted side-by-side between the plier jaws to a depth of approximately 1/2 inch, and shall provide from 7-1/2 to 7-7/8 inches of free wire extending beyond the coupler for twisting. The jaws shall be locked in the gripping position by means of the locking mechanism. The wire shall then be twisted by gripping the spiral rod pull with the thumb and forefinger of one hand and pulling vertically against only the resistance of the weight until the spiral rod reaches its end of travel. The rod shall then be returned to its retracted position and the operation shall be repeated a sufficient number of times to produce a minimum of 10 uniform tight twist of 360 degrees each in the wire. Without releasing the grip on the wire, the pliers shall then be made to lift the weight by means of the twisted wire, and shall hold the suspended weight for a minimum period of 15 seconds.

4.3.2.3 *Small wire twisting test.* The 0.020 inch diameter copper wire tests shall be conducted in the same manner as that specified in 4.3.2.2, except that the seven pound weight shall be substituted for the 35 pound weight, and not less than 16 complete twists shall be made in the wire. In addition, after

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

performing the lifting test, and before removing the twisted wire from the coupler, the grip of the plier jaws shall be released from the ends of the wire, transferred to, and locked on the twisted wire at a point approximately half-way between the free ends and the coupler. The grip shall then be released and transferred back to the ends of the wire, and the lifting test again performed. If the wire breaks at any point it was gripped with the plier jaws, the pliers shall be considered as having excessively weakened the wire.

4.3.3 *Cutting tests.* Each sample wire twister plier shall be subjected to the test of 4.3.3.2, following which the samples shall be subjected to the test of 4.3.3.3.

4.3.3.1 *Test equipment.* The gripping surfaces shall be in a normal horizontal position, with the head resting on a support or guide. The lower handle shall rest on a suitable fixed knife edge support 2/3 the distance from the center of the rivet and the end of the handle. Loads shall be applied through a knife edge to the upper handle 2/3 the distance from the center of the rivet and the end of the handle in line with the lower knife edge support. Test loads applied to the handle shall be calculated by dividing the test load specified herein by the distance between the center of the rivet and the line of action of the load. A suitably equipped platform scale similar to figure 2, a universal testing machine or a specially calibrated testing device employing dead weights may be used.

4.3.3.2 *Wire cutting test.* Each wire twister plier of either the 9-inch or 12-inch size shall cut off three pieces of 0.051 inch diameter Inconel wire specified in 4.3.2. The cuts shall be made at the joint end of the cutting edges and the load required to completely sever the wire shall not exceed 300 inch-pounds. Following completion of the wire cutting test, the pliers shall be subjected to the bond paper cutting test.

4.3.3.3 *Bond paper cutting test.* Paper for this test shall be either rag or chemical wood bond paper. The average thickness of the paper shall not exceed 0.004 inch. Paper shall be in strip form, the width of the strip measuring 1/16 inch less than the length of the cutting edge of the wire twister plier. The strip of paper shall be centered

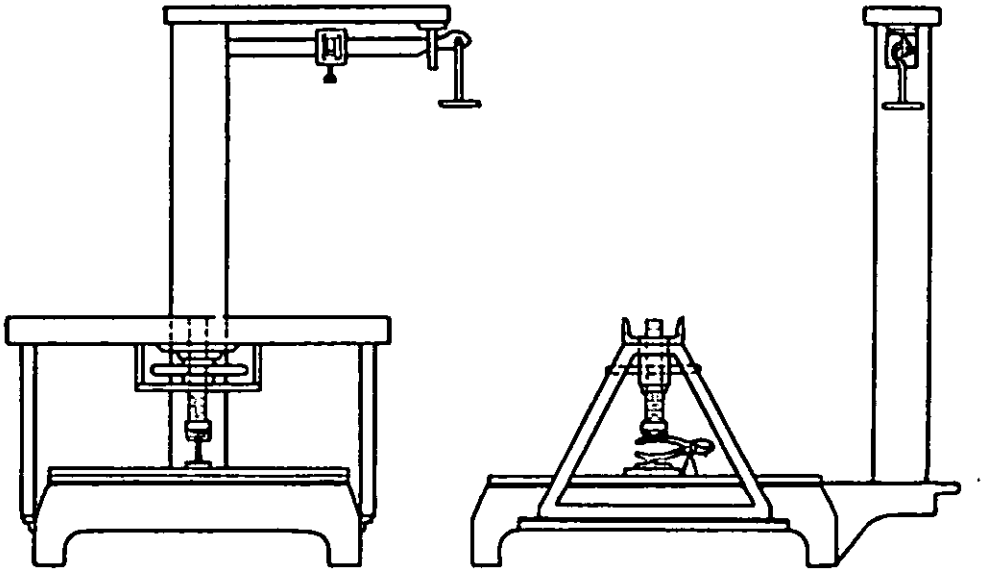


FIGURE 2.—An apparatus for applying test loads to handles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

between the ends of the cutting edges when the cut is being made. The cutters shall completely and cleanly sever the paper when a load not to exceed 75 inch pounds is applied.

4.4 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be examined to determine conformance with the requirements of section 5 of this specification and as required by specifications referenced therein. Except as required by referenced specifications, inspection lots and sampling shall be in accordance with MIL-STD-105, with inspection level L-5 and an AQL of 4.0 percent defective.

5. PREPARATION FOR DELIVERY

(Civil agencies should refer to Fed. Std. No. 102 for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Cleaning, preservation, and packaging.

5.1.1 *Level A.* Level A cleaning, preservation, and packaging shall be in accordance with level A of MIL-H-15424.

5.1.2 *Level C.* Pliers shall be packaged in such a manner that will afford adequate protection against deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This may conform to the supplier's commercial practice, when such meets the requirements of this level.

5.2 Packing.

5.2.1 *Level A.* Level A packing shall be in accordance with level A of MIL-H-15424.

5.2.2 *Level B.* Level B packing shall be in accordance with level B of MIL-H-15424.

5.2.3 *Level C.* Pliers, preserved and packaged as specified in 5.1.1 or 5.1.2 as specified (see 6.1), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Military agencies.* In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with MIL-STD-129.

5.3.2 *Civil agencies.* In addition to any special marking specified in the contract or order, each unit and intermediate package and shipping container shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Size of plier required (see 1.2).

(*c*) Coating, if particular type is required (see 3.4).

(*d*) Whether a definite jaw tip is required (see 3.7).

(*e*) Levels of cleaning, preservation, packaging, and packing required (see section 5).

(*f*) Marking, if different (see 5.3).

6.2 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Tools, not otherwise indexed by name. Carload minimum weight 30,000 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:39Z
Check the source to verify that this is the current version before use.

Motor:

Tools, hand, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.3 Supersession data. The wire twister pliers covered by this specification supersede type XVII of Federal Specification GGG-P-471c.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—QMC

Navy—Ships

Air Force—MOA

☆ U.S. GOVERNMENT PRINTING OFFICE: 1963—633384/283

Copies of this specification may be purchased for 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:39Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:39Z
Check the source to verify that this is the current version before use.